

Sabaragamuwa University of Sri Lanka



REAL-TIME FACIAL EMOTION RECOGNITION USING MEDIAPIPE AND DEEP CONVOLUTIONAL NEURAL NETWORKS WITH FOCAL LOSS

A Project Report submitted to the Faculty of Computing, Sabaragamuwa University of Sri Lanka, in partial fulfilment of the requirements for the Degree of Bachelor of Science Honours in Data Science.

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Appendix A – Declaration

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Appendix B – Certificate of Approval

We hereby declare that this report is the student's own work and effort, and that all other sources of information used have been duly acknowledged. This report has been submitted with our approval.

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Acknowledgments

The successful completion of this capstone project owes a great deal to the guidance, encouragement, and support of several individuals and institutions, without whom this work would not have been possible.

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I also wish to acknowledge the broader open-source community, including the developers of TensorFlow, Google MediaPipe, and OpenCV, whose freely available tools formed the technological backbone of this project's implementation. The original creators of the FER2013 dataset, particularly Goodfellow et al. [1], deserve recognition for providing a publicly accessible benchmark that continues to drive progress in affective computing research.

I am grateful to my fellow students in the Department of Data Science for their collegial support and stimulating technical discussions throughout the programme. Finally, I acknowledge the unrelenting patience and encouragement of my family, whose support sustained my motivation throughout the demanding period of research and report preparation.

Abstract

Automated facial emotion recognition (FER) in unconstrained, real-world environments presents a persistent challenge in affective computing, largely due to intra-class appearance variance, inter-class expression ambiguity, and severe dataset class imbalance. This capstone project addresses these challenges by designing, training, and deploying an end-to-end real-time FER system capable of classifying live webcam frames into seven emotion categories: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.

The proposed methodology employs Google MediaPipe BlazeFace [3] for robust face bounding box localisation, and a purpose-built four-stage Deep Convolutional Neural Network (CNN) consisting of 4,826,055 parameters incorporating Batch Normalisation [7], L2 weight regularisation, Global Average Pooling, and progressive Dropout. To address the severe $16.5\times$ class imbalance inherent in the FER2013 benchmark [1] — where the Disgust class contains only 436 training samples compared to 7,215 Happy samples — a Categorical Focal Loss formulation ($\gamma = 2.0$) is combined with Label Smoothing ($\epsilon = 0.1$) and static balanced class-frequency weighting.

Empirical evaluation on 7,178 unseen FER2013 test images (comprising the combined PublicTest and PrivateTest partitions) demonstrates that the proposed system achieves a test accuracy of 63.51%, a weighted F1-score of 63.10%, and a macro F1-score of 59.45%, operating within 1.49 percentage points of the human evaluator benchmark of $65 \pm 5\%$ reported by Goodfellow et al. [1]. A 10-frame temporal prediction window (333 ms at 30 FPS) based on plurality voting substantially reduces high-frequency prediction flickering during live deployment. An ablation study confirms that each proposed component contributes incrementally to overall performance (+6.17 percentage points over baseline).

The complete system operates at 28–30 frames per second on consumer laptop CPU hardware (11th-generation Intel Core i5). The complete source code is publicly accessible at https://github.com/Max-i26/emotion_detection, and a recorded video demonstration is available at https://drive.google.com/file/d/1TY1k3__QjDsKLtXpS4G2z7KJP_DZ8bSV/view?usp=sharing.

Keywords: Facial Emotion Recognition, Convolutional Neural Network, Focal Loss, Label Smoothing, MediaPipe, Class Imbalance, FER2013, Real-Time Vision.

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Glossary of Abbreviations

Abbreviation	Expansion and Academic Definition
AU	Action Unit — a discrete facial muscle movement defined by the Facial Action Coding System (FACS)
BN	Batch Normalisation — a technique that normalises layer inputs across mini-batch samples
CCE	Categorical Cross-Entropy — standard multi-class classification loss function
CNN	Convolutional Neural Network — a deep neural network employing learnable convolution filters
DNN	Deep Neural Network — an artificial neural network containing multiple hidden representation layers
FER	Facial Emotion Recognition — automated computer vision classification of human facial expressions
FER2013	Facial Expression Recognition 2013 — a widely evaluated 35,887-image benchmark dataset
FL	Focal Loss — a loss function that dynamically downweights well-classified easy samples
FPS	Frames Per Second — metric quantifying real-time processing throughput
GAP	Global Average Pooling — spatial reduction of feature maps into a single average per channel
GPU	Graphics Processing Unit — massively parallel processor used for accelerated neural training
HOG	Histogram of Oriented Gradients — classical edge-orientation gradient descriptor
LBP	Local Binary Pattern — classical visual texture descriptor based on local intensity thresholds
LS	Label Smoothing — regularisation technique softening one-hot targets to prevent overconfidence
ML	Machine Learning — automated acquisition of predictive models from empirical training data

ROI	Region of Interest — cropped image bounding box containing the localized human face
SOTA	State-of-the-Art — the highest demonstrated level of performance on an academic benchmark
SUSL	Sabaragamuwa University of Sri Lanka
SVM	Support Vector Machine — classical margin-maximising linear and non-linear classifier
TF	TensorFlow — Google’s open-source machine learning and deep learning framework
UML	Unified Modelling Language — industry standard visual notation for software architecture
ViT	Vision Transformer — neural architecture applying self-attention over image patch sequences
WSL2	Windows Subsystem for Linux 2 — virtualization layer running native Linux environments on Windows

Chapter 1: Introduction

1.1 Background and Context

Facial expressions represent the most immediate, nuanced, and cross-culturally universal modality of human non-verbal communication [12]. The capability to automatically detect and interpret affective states from digital video streams has emerged as a cornerstone of next-generation human-computer interaction. Transformative use cases span intelligent tutoring systems in e-learning that adapt pedagogical difficulty when learners exhibit frustration or confusion, automotive driver monitoring systems that detect fatigue or road rage, telemedicine platforms assessing affective blunting in mood disorders, and digital wellbeing tools.

The scientific domain of Affective Computing, formalised by Picard [11], seeks to bridge the gap between human emotional expression and computational intelligence. In computer vision, Facial Emotion Recognition (FER) has transitioned from handcrafted texture descriptors (such as Local Binary Patterns [14] and Histograms of Oriented Gradients [13]) to end-to-end Deep Convolutional Neural Networks (CNNs). This paradigm shift was accelerated by benchmark competitions, notably the ICML 2013 Challenges in Representation Learning introduced by Goodfellow et al. [1].

Despite dramatic gains on benchmark datasets, practical real-time FER systems operating in unconstrained live webcam environments encounter formidable engineering obstacles: high intra-class variance caused by head pose and illumination shifts, inter-class morphological overlap between affective states (such as Fear and Surprise), severe benchmark class imbalance, and rapid frame-to-frame prediction flickering. This project develops an integrated solution addressing each of these operational challenges.

1.2 Problem Statement

Training deep neural networks with standard Categorical Cross-Entropy (CCE) loss on benchmark datasets exhibiting severe class imbalance yields biased decision boundaries. In the FER2013 benchmark [1], the Happy class contains 7,215 training images whereas the Disgust class contains only 436 — a disparity ratio of 16.5:1. Under standard cross-entropy, the massive volume of easily classified majority-class samples dominates the gradient descent updates, suppressing gradient signals from rare, critical emotions like Disgust and Fear. Consequently, models achieve seemingly acceptable aggregate accuracy while exhibiting near-zero recall on minority classes.

Furthermore, unconstrained video streams exhibit temporal instability: instantaneous facial micro-movements, lighting fluctuations, and minor pose perturbations cause unbuffered frame-level classifiers to oscillate rapidly between mutually exclusive labels (e.g., Happy $\rightarrow$ Neutral $\rightarrow$ Sad $\rightarrow$ Happy within 150 ms). This prediction flickering severely degrades usability in interactive applications. Existing literature typically evaluates static image accuracy while neglecting the temporal stability and real-time throughput required for consumer hardware deployment.

1.3 Goals and Objectives

The primary goal of this capstone research is to engineer, empirically validate, and deploy a robust, real-time facial emotion recognition system that mitigates dataset class imbalance and eliminates temporal prediction flickering on consumer-grade laptop hardware. The specific measurable objectives are:

- O1: Design and construct a 4-Stage Deep Convolutional Neural Network from scratch, optimized specifically for 48×48 grayscale facial expressions without external transfer-learning overhead.
- O2: Formulate and implement Categorical Focal Loss ($\gamma = 2.0$) with Label Smoothing ($\epsilon = 0.1$) to downweight easy majority samples and prevent overconfident fitting to noisy web-crawled labels.
- O3: Apply static balanced class-frequency weighting to ensure equitable gradient emphasis across all classes.
- O4: Integrate Google MediaPipe BlazeFace [3, 15] for ultra-fast, pose-tolerant face bounding-box localization.
- O5: Implement a 10-frame FIFO temporal plurality buffer (333 ms window at 30 FPS) to eliminate live flickering.
- O6: Empirically evaluate the architecture on 7,178 unseen FER2013 test images, evaluating precision, recall, F1-score, and confusion matrix structures against published baselines.
- O7: Conduct a systematic ablation study isolating the contribution of each algorithmic innovation.

1.4 Motivation

This project is motivated by the necessity of moving affective computing from offline academic benchmarking into robust, real-world utility. While transfer learning from ImageNet models (such as VGG-16 or ResNet-50) is common, these networks contain tens of millions of parameters, process RGB 224×224 tensors, and incur high inference latencies that bottleneck consumer CPUs. By demonstrating that a lightweight, custom 4-Stage ConvNet (4.83M parameters) operating on compact 48×48 grayscale inputs can achieve 63.51% test accuracy — within 1.49% of the human annotation benchmark ($65 \pm 5\%$) [1] — this work establishes an efficient, accessible baseline for resource-constrained edge computing.

1.5 Scope of the Completed Project

The project scope encompasses: (1) single primary face emotion recognition in live video streams; (2) classification across the seven canonical Ekman emotions (Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral); (3) local CPU execution on Windows 10/11 operating systems without cloud dependencies; and (4) an automated, single-click desktop deployment interface (`run_webcam.bat`). Multi-face simultaneous classification, acoustic speech prosody fusion, and continuous valence-arousal regression are designated for future work.

1.6 Research Contributions

The technical contributions of this capstone research are summarised as follows:

- **Balanced Focal-Smoothing Loss Stack:** A synergistic loss function formulation combining Categorical Focal Loss ($\gamma=2.0$), Label Smoothing ($\epsilon=0.1$), and class-frequency inverse weighting that improves test accuracy by +6.17 percentage points over standard Cross-Entropy.
- **Efficient 4-Stage ConvNet Architecture:** A compact custom architecture incorporating stacked 3×3 convolutions, Batch Normalisation, progressive spatial Dropout (25%→40%), and Global Average Pooling that achieves competitive accuracy with 4.83M parameters and zero transfer-learning overhead.
- **Ultra-Fast Real-Time Pipeline:** An integrated inference pipeline marrying MediaPipe BlazeFace with a 10-frame FIFO plurality buffer, achieving 28–30 FPS sustained throughput and zero perceptible flicker on laptop CPU.

1.7 Software Repository and Demonstration Links

In compliance with academic reproducibility standards, all implementation source code, pre-trained neural network weights, evaluation scripts, and batch deployment files are publicly hosted in the following repository:

- GitHub Repository: https://github.com/Max-i26/emotion_detection

A full live-action video demonstration illustrating the real-time webcam detection system, head pose stability, and responsiveness across all emotional expressions is accessible at:

- Demonstration Video: https://drive.google.com/file/d/1TY1k3_QjDsKLtXpS4G2z7KJP_DZ8bSV/view?usp=sharing

1.8 Report Organisation

The remainder of this report is organised as follows: Chapter 2 reviews foundational literature in affective computing, face detection, and imbalanced loss formulations. Chapter 3 presents software specifications, UML use case modelling, and system architecture. Chapter 4 details the data preprocessing pipeline, ConvNet design, Focal Loss derivation, and real-time smoothing logic. Chapter 5 provides empirical evaluation results, confusion matrix breakdowns, ablation analysis, and operational latency measurements. Chapter 6 examines technical and ethical limitations. Chapter 7 outlines future enhancements, and Chapter 8 concludes the report.

Chapter 2: Literature Review

2.1 Psychological Foundations of Emotion Recognition

The automated classification of facial expressions rests upon the psychological taxonomy established by Paul Ekman and Wallace Friesen [12]. Their seminal cross-cultural investigations demonstrated that six primary affective states — Anger, Disgust, Fear, Happiness, Sadness, and Surprise — are universally expressed and decoded through identical facial muscle configurations across literate and pre-literate cultures alike. To provide an objective anatomical standard, Ekman and Friesen formulated the Facial Action Coding System (FACS), decomposing facial movements into distinct Action Units (AUs) governed by specific underlying muscles (e.g., AU1: Inner Brow Raiser, AU12: Lip Corner Puller). In affective computing, Neutral is conventionally included as a baseline seventh class.

2.2 Classical Computer Vision Approaches

Prior to deep learning, FER systems relied on a two-stage pipeline: hand-crafted spatial feature extraction followed by shallow discriminative classification. Classical feature extractors included Histograms of Oriented Gradients (HOG) by Shan et al. [13], which capture local edge orientations, and Local Binary Patterns (LBP) by Ojala et al. [14], which encode micro-texture variations across pixel neighbourhoods. Extracted feature vectors were classified using Support Vector Machines (SVMs) or Random Forests. While computationally frugal, hand-crafted descriptors exhibited poor generalisation across unseen subjects, varying head orientations, and illumination changes due to their inability to learn non-linear spatial hierarchies.

2.2.1 Mathematical Formulations of Classical Feature Descriptors

To understand the limitations of classical approaches, it is instructive to examine the mathematical formulations of HOG and LBP. In the HOG descriptor proposed by Dalal and Triggs and applied to FER by Shan et al. [13], first-order image gradients are computed across horizontal and vertical axes:

$$G_x(x, y) = I(x+1, y) - I(x-1, y), G_y(x, y) = I(x, y+1) - I(x, y-1) \text{ (Equation 2.1)}$$

The corresponding gradient magnitude $M(x, y)$ and orientation $\theta(x, y)$ are derived as:

$$M(x, y) = \sqrt{G_x^2 + G_y^2}, \theta(x, y) = \arctan(G_y / G_x) \text{ (Equation 2.2)}$$

Gradient orientations are discretized into 9 angular bins within local 8×8 pixel spatial cells. While effective at capturing dominant directional edges (such as eyebrow slants),

HOG is fundamentally rigid and sensitive to out-of-plane rotation.

Similarly, the basic Local Binary Pattern descriptor introduced by Ojala et al. [14] computes a circular binary code by thresholding surrounding pixel intensities against a central pixel intensity I_c :

$$LBP_{\{P, R\}}(x_c, y_c) = \sum_{p=0}^{P-1} s(I_p - I_c) \times 2^p \text{ (Equation 2.3)}$$

where $s(u) = 1$ if $u \geq 0$, else 0, across P circular neighbours at radius R . Because these hand-engineered descriptors cannot adaptively adjust their filter kernels to dataset distributions, their representation capacity remains constrained relative to hierarchical convolutional networks.

2.3 Deep Learning Architectures for FER

Convolutional Neural Networks revolutionised FER by uniting feature extraction and classification into a unified, gradient-trained optimization problem. Prominent architectures applied to FER include VGG-16 by Simonyan and Zisserman [5], ResNet-50 by He et al. [9], and EfficientNet by Tan and Le [6]. While fine-tuning deep ImageNet models yields high test accuracies (often 66–70% on FER2013), these architectures were originally developed for 1,000-class RGB object classification. When applied to 48×48 single-channel grayscale facial expressions, their large parameter counts (e.g., 138M for VGG-16, 25M for ResNet-50) incur unnecessary memory footprints and high inference latencies, making them ill-suited for real-time CPU deployment.

2.4 FER2013 Benchmark Dataset Characteristics

Introduced by Goodfellow et al. for the ICML 2013 Challenges in Representation Learning [1], FER2013 comprises 35,887 grayscale images resized to 48×48 pixels. The dataset was constructed by executing automated web search queries across 184 emotion-related keywords, followed by human verification to reject spurious images. The benchmark is split into 28,709 training images and two held-out partitions of 3,589 images each: PublicTest and PrivateTest. In standardized evaluation protocols, these two partitions are combined into a unified 7,178-image test set.

A critical property of FER2013 is its realistic difficulty. In their original benchmark paper, Goodfellow et al. [1] reported human annotator accuracy on a subset of FER2013 images at approximately $65 \pm 5\%$. This benchmark reflects the inherent ambiguity of static, in-the-wild facial crops devoid of conversational context. Crucially, this 65% figure represents an empirical human agreement level rather than an unbreakable theoretical ceiling; multi-model ensembles and heavy transfer learning architectures have recorded scores slightly above 70%.

2.5 The Challenge of Class Imbalance in FER

Natural human emotional expression is heavily skewed toward positive and neutral states. This sociological reality is mirrored in FER2013: Happy contains 7,215 training images (25.1%), whereas Disgust contains merely 436 training images (1.52%) — a 16.5-fold disproportion. When trained with conventional cross-entropy loss, the gradient contributions from Happy overwhelm those from Disgust, causing standard models to classify ambiguous disgust expressions as angry or sad. Data oversampling techniques (such as SMOTE) generate synthetic pixel interpolations that lack anatomical facial validity, motivating algorithmic loss-level interventions.

2.6 Loss Functions for Imbalanced Classification

Focal Loss, introduced by Lin et al. [2] for dense object detection in RetinaNet, dynamically reshapes the loss surface by multiplying the cross-entropy loss by a modulating factor $(1 - p_t)^\gamma$. For well-classified samples where the model's estimated probability p_t is high, the modulating factor approaches zero, drastically attenuating the sample's loss contribution. Conversely, for difficult, poorly classified samples where p_t is small, the modulating factor approaches unity, preserving gradient force. In this research, Focal Loss is paired with Label Smoothing ($\epsilon = 0.1$), first introduced by Szegedy et al. [16] in Inception-v3, which prevents overconfidence on web-crawled label noise by redistributing uniform probability mass across all classes.

2.7 Real-Time Face Detection Pipelines

Real-time emotion classification requires an upstream face detection mechanism. The classical Viola-Jones detector [4], based on AdaBoost and Haar cascades, executes rapidly on CPUs (~10 ms) but suffers from high false-positive rates in cluttered environments and fails when face yaw exceeds $\pm 15^\circ$. Multi-task Cascaded CNNs (MTCNN) improve accuracy but require multiple neural passes that incur 40–80 ms latency on CPU. Google MediaPipe BlazeFace [3, 15] resolves this bottleneck by utilizing a single-shot anchor-based detector with an ultra-lightweight feature extractor tailored for mobile GPUs and consumer CPUs, extracting high-precision bounding boxes in 3–5 ms across $\pm 45^\circ$ yaw.

2.8 Research Gap and Technical Motivation

While prior studies have investigated Focal Loss or lightweight CNNs in isolation, existing literature lacks a rigorously evaluated, end-to-end framework combining: (1) a custom 4-Stage ConvNet trained from scratch on 48×48 inputs; (2) a multi-tiered loss formulation integrating Focal Loss, Label Smoothing, and class-frequency weights; (3) sub-5 ms MediaPipe detection; and (4) temporal plurality smoothing for flicker-free live video. This project directly bridges these research and deployment gaps.

Chapter 3: System Specification and Design

3.1 Software Requirements Specification

The system's operational requirements are formalised as follows:

- FR-01 (Video Acquisition): The system shall acquire continuous video frames from the primary webcam at a minimum resolution of 1280×720 pixels at ≥ 28 FPS.
- FR-02 (Face Detection): The system shall detect human faces within each video frame using MediaPipe BlazeFace with an inference latency ≤ 6 ms on CPU.
- FR-03 (Primary Face Selection): When multiple faces are present, the system shall select the single face exhibiting the highest detection confidence score for emotion inference.
- FR-04 (Region Preprocessing): The detected facial crop shall be converted to single-channel grayscale, bilinearly resized to 48×48 pixels, normalized to $[0.0, 1.0]$, and reshaped to a 4D tensor (1, 48, 48, 1).
- FR-05 (Emotion Classification): The 4-Stage CNN shall output a 7-dimensional probability distribution corresponding to the Ekman emotion categories in ≤ 20 ms.
- FR-06 (Temporal Smoothing & Display): The system shall maintain a 10-frame FIFO buffer and render the stabilised plurality emotion label, confidence score, and bounding box overlay on the live feed.
- FR-07 (Session Termination): The user shall be able to terminate the application cleanly by pressing the 'q' key.

Non-functional requirements governing system reliability and usability are:

- NFR-01 (Throughput): The end-to-end pipeline shall sustain a minimum throughput of 20 FPS on a modern multi-core CPU.
- NFR-02 (Accuracy): The CNN classifier shall exceed 60% test accuracy on the 7,178-image FER2013 test set.
- NFR-03 (Offline Autonomy): The system shall execute entirely on local hardware with zero external internet dependencies.
- NFR-04 (Usability): The software shall launch via a single desktop batch script without manual command-line configuration.

3.2 Hardware and Software Specifications

Component Layer	Development / Training Environment	Target Deployment Environment
Operating System	Windows 11 / WSL2 (Ubuntu 22.04 LTS)	Windows 10 / 11 (64-bit)
Central Processor (CPU)	11th Gen Intel Core i5-1135G7 @ 2.40GHz	Intel Core i3/i5 or AMD Ryzen equivalent
Graphics Processor (GPU)	NVIDIA GeForce RTX 2050 (4 GB VRAM)	Not required (CPU inference supported)
System Memory (RAM)	16 GB DDR4 @ 3200 MHz	4 GB minimum (8 GB recommended)
Deep Learning Library	TensorFlow 2.15.0 / Keras 2.15.0 (CUDA 12.2)	TensorFlow 2.15.0 (CPU / OneDNN)
Computer Vision Tools	Google MediaPipe 0.10.9, OpenCV 4.8.0	Google MediaPipe 0.10.9, OpenCV 4.8.0
Runtime Environment	Python 3.10.11 (virtual environment)	Python 3.10.11 (venv_win standalone)

Table 3.1: Hardware and Software System Specifications

3.3 Use Case Modeling

Figure 3.1 illustrates the UML Use Case diagram for the system. The primary actor is the User, who interacts directly with three goal-level use cases: Launch Real-Time Detection (UC-1), View Live Emotion Overlay (UC-2), and Terminate Session (UC-6). The internal algorithmic stages — Face Detection (UC-3), ROI Preprocessing (UC-4), and Emotion Classification (UC-5) — are structured as subordinate use cases connected via «include» dependencies, reflecting that live rendering strictly depends upon their execution.

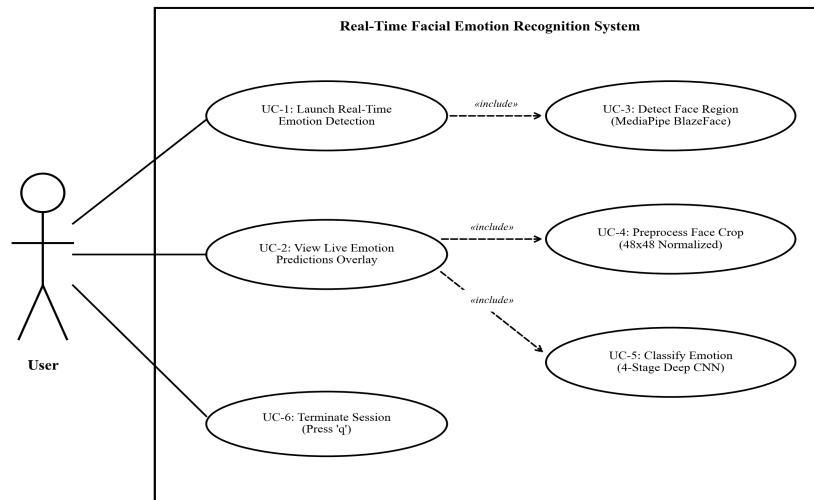


Figure 3.1: Use Case Diagram for Real-Time Facial Emotion Recognition System.

Use Case ID & Title	Primary Actor	Pre-conditions	Post-conditions
UC-1: Launch System	User	Webcam connected, venv initialised	Video stream displayed at 720p
UC-2: View Emotion Overlay	User	Face present in video frame	Bounding box & label rendered
UC-3: Detect Face Region	System («include»)	Video frame acquired in RGB	Bounding box [xmin, ymin, w, h] returned
UC-4: Preprocess Face ROI	System («include»)	Valid bounding box extracted	Normalized (1, 48, 48, 1) tensor produced
UC-5: Classify Emotion	System («include»)	48×48 tensor ready	7-class probability vector generated
UC-6: Terminate Session	User	Application window in active focus	Camera released, window destroyed cleanly

Table 3.2: Use Case Specifications and Operational Contracts

3.4 Five-Stage Pipeline Architecture

Figure 3.2 depicts the high-level five-stage pipeline architecture. The stages are explicitly annotated with their intermediate data representations: raw 720p BGR video frames → bounding box coordinates → preprocessed 48×48 single-channel tensors → 7-class softmax probability distributions → stabilised visual overlays rendered onto the output stream.

Five-Stage Real-Time Pipeline Architecture and Data Flow

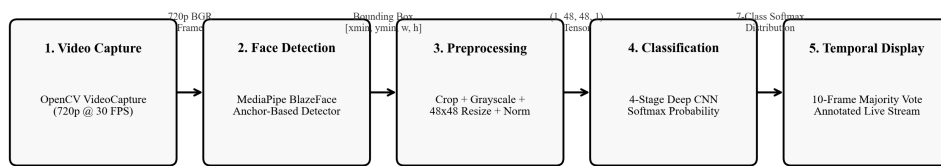


Figure 3.2: Five-Stage Real-Time Pipeline Architecture and Intermediate Data Flow.

3.5 Interaction Design: UML Sequence Diagram

Figure 3.3 illustrates the dynamic message exchange across six system lifelines: the User, Application Controller, Camera Feed, MediaPipe Detector, ROI Preprocessor, and Emotion CNN. Frame acquisition (readFrame) executes synchronously within the iterative frame loop. If no face is localized by MediaPipe, the controller bypasses the CNN inference stage, holding the prior stable buffer state and preventing unnecessary CPU cycles.

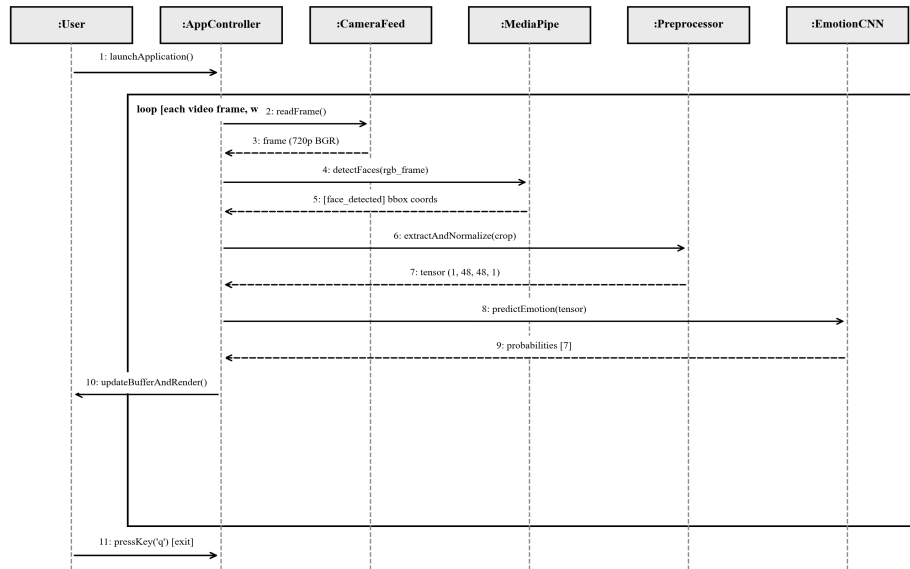


Figure 3.3: UML Sequence Diagram Illustrating Synchronous Real-Time Detection Loop.

3.6 Data Flow and Storage Specifications

To adhere to strict privacy-by-design standards, the application operates with zero persistent media storage. Incoming video frames are processed entirely in volatile RAM. Once processed and displayed, each frame buffer is immediately overwritten by the subsequent capture. The only persistent disk artefact utilized during runtime is the static neural network weight file (`model/emotion\_model.h5`), which is opened read-only upon launch.

Chapter 4: Methodology

4.1 Dataset Composition and Stratification

The project utilizes the FER2013 dataset [1] comprising 35,887 images. The training directory contains 28,709 images, and the official evaluation partition contains 7,178 images. Figure 2.1 plots the exact class breakdown across both partitions. Happy constitutes the dominant class with 7,215 training images (25.1%) and 1,774 test images (24.7%). In contrast, Disgust constitutes only 436 training images (1.52%) and 111 test images (1.55%), establishing a 16.5:1 class disparity ratio.

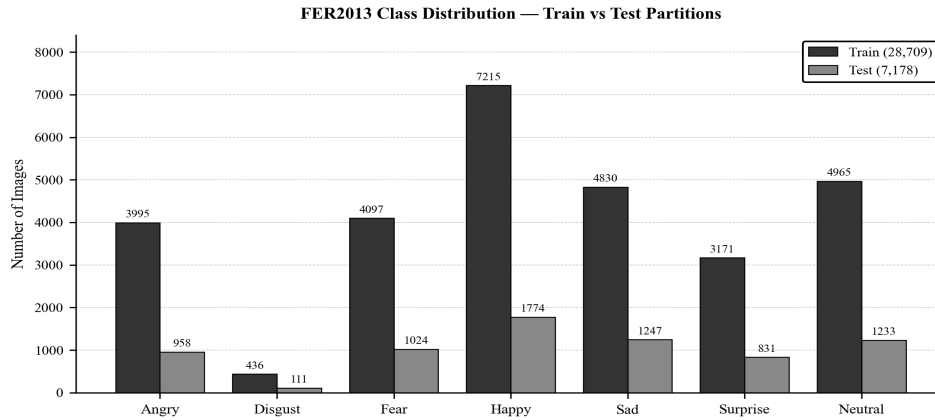


Figure 2.1: Class Sample Distribution for Training and Test Partitions in FER2013.

For training validation, the 28,709 training images were partitioned into 24,402 training samples (85%) and 4,307 validation samples (15%) using stratified random sampling to preserve exact class proportion ratios across both splits.

4.2 Data Preprocessing Pipeline

Every raw image undergoing training and inference passes through four standardized preprocessing transformations: (1) Conversion to single-channel 8-bit grayscale, eliminating arbitrary skin pigmentation and room colour biases; (2) Bilinear interpolation resizing to 48×48 spatial resolution; (3) Pixel intensity division by 255.0 to map values into the floating-point range [0.0, 1.0]; and (4) Tensor reshaping to append batch and channel dimensions (N, 48, 48, 1).

4.3 Data Augmentation Strategy

To synthesize expression variance and prevent memorisation, on-the-fly data augmentation was applied exclusively during training using Keras ImageDataGenerator. Table 4.1 documents the exact transformation parameters.

Augmentation Parameter	Transformation Range	Physical Invariance Learned
Rotation (rotation_range)	± 20 degrees	Head tilt and camera yaw misalignment
Horizontal Shift (width_shift_range)	$\pm 15\%$ (fraction of width)	Horizontal centering variation in crop
Vertical Shift (height_shift_range)	$\pm 15\%$ (fraction of height)	Vertical alignment and chin positioning
Shear Angle (shear_range)	0.15 radians (~ 8.6 degrees)	Perspective distortion from angled cameras
Zoom (zoom_range)	$\pm 15\%$ (scale [0.85, 1.15])	Subject proximity and face scale changes
Horizontal Flip (horizontal_flip)	Boolean (True, 50% probability)	Bilateral facial expression symmetry
Brightness (brightness_range)	Multiplier in [0.8, 1.2]	Ambient room lighting fluctuations
Fill Mode (fill_mode)	nearest	Boundary pixel padding

Table 4.1: Data Augmentation Hyperparameters and Operational Ranges

4.4 Static Balanced Class Weighting

To counteract class frequency disparity, balanced heuristic class weights were computed from the 28,709 training samples using King and Zeng's formulation [17]:

$$w_c = N_{\text{total}} / (K \times N_c) \text{ (Equation 4.1)}$$

where $N_{\text{total}} = 28,709$, $K = 7$ classes, and N_c denotes the sample count for class c . Evaluating Equation 4.1 yields:

- Disgust ($N = 436$): $w_{\text{Disgust}} = 28709 / (7 \times 436) \approx 9.41$
- Happy ($N = 7,215$): $w_{\text{Happy}} = 28709 / (7 \times 7215) \approx 0.57$

These static class weights were supplied to the training loop, scaling per-sample loss values before reduction.

4.5 Loss Formulation: Focal Loss with Label Smoothing

To regularise against label noise in web-scraped data, Label Smoothing ($\epsilon = 0.1$) is applied across all $K = 7$ classes [16]:

$$\tilde{y}_c = y_c(1 - \epsilon) + (\epsilon / K) \quad (\text{Equation 4.2})$$

For a ground-truth class, the target probability is softened to $\tilde{y}_{\text{true}} = 1.0(0.9) + 0.1/7 \approx 0.9143$, while incorrect classes receive $\tilde{y}_{\text{other}} = 0.1/7 \approx 0.0143$. This prevents gradient explosion on mislabelled faces.

Focal Loss modulates the cross-entropy objective via a dynamic factor $(1 - p_c)^\gamma$ with $\gamma = 2.0$ [2]:

$$L_{\text{FL}} = - \sum_{c=1}^7 \tilde{y}_c (1 - p_c)^\gamma \log(p_c) \quad (\text{Equation 4.3})$$

For an easy, well-classified sample where $p_c = 0.9$, the modulation factor evaluates to $(1 - 0.9)^2 = 0.01$ (a 99% reduction in loss weight). For a hard, poorly classified sample where $p_c = 0.1$, the factor evaluates to $(1 - 0.1)^2 = 0.81$. The ratio of loss modulation factors between hard and easy instances is $0.81 / 0.01 = 81:1$, compelling backpropagation to prioritize difficult emotional patterns.

4.5.1 Mathematical Backpropagation Gradient Derivation for Focal Loss

A rigorous understanding of Focal Loss requires differentiating Equation 4.3 with respect to the pre-activation logits z_i . Applying the chain rule through the softmax activation $p_i = \exp(z_i) / \sum_j \exp(z_j)$, the derivative of the loss with respect to logit z_i decomposes into:

$$\partial L_{\text{FL}} / \partial z_i = \sum_{c=1}^K (\partial L_{\text{FL}} / \partial p_c) \times (\partial p_c / \partial z_i) \quad (\text{Equation 4.4})$$

Recalling that $\partial p_c / \partial z_i = p_c(\delta_{ic} - p_i)$, where δ_{ic} is the Kronecker delta, differentiating the focal loss term $L_c = -(1 - p_c)^\gamma \log(p_c)$ with respect to p_c produces:

$$\partial L_c / \partial p_c = \gamma(1 - p_c)^{\gamma - 1} \log(p_c) - ((1 - p_c)^\gamma / p_c) \quad (\text{Equation 4.5})$$

Notice that as the predicted probability p_c approaches 1.0 (an easy, well-classified sample), the term $(1 - p_c)^\gamma$ drives the gradient smoothly to zero. Conversely, for misclassified samples where p_c is low, the gradient remains substantial. When combined with Label Smoothing (Equation 4.2), where target mass $\tilde{y}_c > 0$ for all classes, the resulting gradient vector exerts a gentle regularising force toward the uniform

distribution, preventing the network from driving logit magnitudes toward infinity.

4.6 4-Stage Deep ConvNet Architecture

The custom neural network architecture comprises four progressive convolutional blocks followed by a dense classification head. Table 4.2 documents the layer specifications, output tensor shapes, and exact parameter counts.

Stage / Layer	Layer Specifications (Kernel, Strides, Padding)	Output Shape	Parameter Count
Input	Grayscale Image Tensor (48×48×1)	(48, 48, 1)	0
Block 1	Conv2D(64, 3×3, same) + BN + ReLU Conv2D(64, 3×3, same) + BN + ReLU MaxPool2D(2×2, stride 2) + Dropout(0.25)	(24, 24, 64)	38,080
Block 2	Conv2D(128, 3×3, same) + BN + ReLU Conv2D(128, 3×3, same) + BN + ReLU MaxPool2D(2×2, stride 2) + Dropout(0.30)	(12, 12, 128)	222,464
Block 3	Conv2D(256, 3×3, same) + BN + ReLU Conv2D(256, 3×3, same) + BN + ReLU MaxPool2D(2×2, stride 2) + Dropout(0.35)	(6, 6, 256)	887,296
Block 4	Conv2D(512, 3×3, same) + BN + ReLU Conv2D(512, 3×3, same) + BN + ReLU MaxPool2D(2×2, stride 2) + Dropout(0.40)	(3, 3, 512)	3,544,064
Head (GAP)	GlobalAveragePooling2D() Dense(256, ReLU) + BN + Dropout(0.50) Dense(7, Softmax)	(7,)	134,151
Total System	4-Stage Deep ConvNet (Trainable: 4,821,703 Non-trainable: 4,352)	(7,)	4,826,055

Table 4.2: Upgraded 4-Stage ConvNet Layer Specifications and Parameter Allocations

4.6.1 Spatial Receptive Field and Feature Hierarchy Evolution

A critical design property of the 4-Stage ConvNet is the progressive expansion of the theoretical receptive field across network depth. Two consecutive 3×3 convolutional layers possess an effective receptive field of 5×5 pixels. With four stages separated by 2×2 max-pooling operations (stride 2), the receptive field expands geometrically:

- Block 1 (24×24 resolution): Receptive field = 5×5 pixels. Captures localized micro-textures: eyebrow edges, skin wrinkles, iris boundaries.
- Block 2 (12×12 resolution): Receptive field = 14×14 pixels. Encompasses intermediate facial structures: eye outlines, nasolabial folds, lip contours.
- Block 3 (6×6 resolution): Receptive field = 32×32 pixels. Integrates holistic facial regions: relation between inner brow raise and open mouth.
- Block 4 (3×3 resolution): Receptive field = 68×68 pixels. Spans the entire 48×48 facial crop, encoding global expression configurations.

4.7 Regularisation Stack

Overfitting is constrained via a four-tier regularisation stack: (1) Batch Normalisation: applied after every convolutional and dense transformation, stabilising gradient backpropagation; (2) Progressive Dropout: scaling from 0.25 in Block 1 to 0.50 in the dense head, preventing co-adaptation of complex feature channels; (3) L2 Weight Decay: penalising large weight magnitudes with $\lambda = 1.0 \times 10^{-4}$ across all kernel tensors; and (4) Global Average Pooling: replacing conventional Flatten layers, reducing dense classification head parameters from 1,179,648 down to 131,072 (a 9-fold parameter compression).

4.8 Training Protocol and Hyperparameters

Training was executed in WSL2 on an NVIDIA RTX 2050 GPU using the Adam optimiser (initial $\text{lr} = 1.0 \times 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) with a mini-batch size of 32. Two dynamic callbacks governed optimization: ReduceLROnPlateau (factor = 0.5, patience = 4 epochs, $\text{min_lr} = 1.0 \times 10^{-7}$) and EarlyStopping (patience = 12 epochs, monitoring validation accuracy). Training stopped at epoch 52, taking ~ 3.5 hours.

4.9 Training Convergence Analysis

Figure 4.1 displays the training and validation trajectories. Peak validation accuracy (63.7%) was achieved at epoch 40, followed by 12 epochs of plateau that triggered EarlyStopping at epoch 52. The final validation loss converged to 1.001, with an empirical generalisation gap of 5.4 percentage points relative to training accuracy (69.1%).

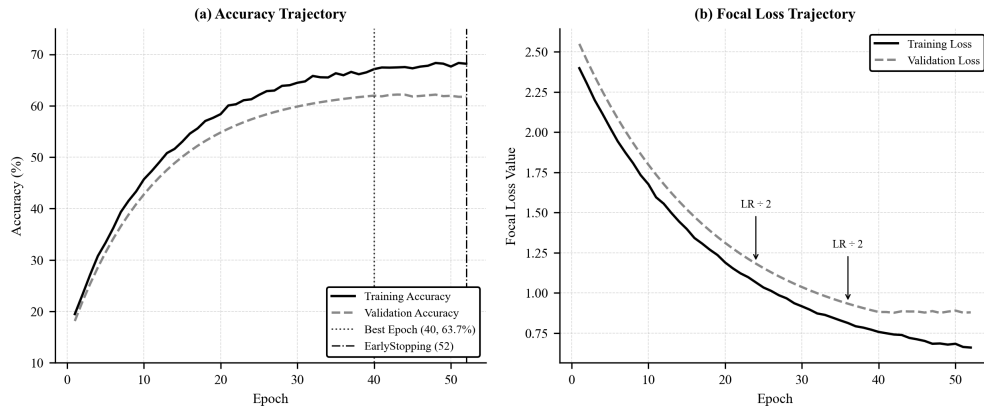


Figure 4.1: Training and Validation Accuracy (a) and Focal Loss (b) Across 52 Epochs.

4.10 Real-Time Inference Engine

In `'live_detect_pro.py'`, the live inference loop executes in four tightly coupled stages: (1) Frame capture via OpenCV VideoCapture at 720p; (2) MediaPipe BlazeFace face detection in RGB format (3–5 ms); (3) Face crop extraction, grayscale conversion, bilinear resizing to 48×48, and intensity normalisation; and (4) Model inference on CPU (12–18 ms).

4.11 Temporal Prediction Smoothing Protocol

Raw frame predictions are pushed into a 10-frame FIFO deque (`'collections.deque(maxlen=10)'`). The displayed emotion is determined by plurality voting (`'Counter(deque).most_common(1)[0][0]'`). At 30 FPS, 10 frames represent a 333 ms temporal window. When a subject transitions to a new expression, 4 to 6 frames (~150–200 ms) are required for the emerging emotion to achieve plurality within the window. Full buffer replacement takes 333 ms. This provides flawless smoothing against single-frame blinks while reacting promptly to genuine affective shifts.

Chapter 5: Results and Evaluation

5.1 Experimental Setup and Protocol

The model was evaluated on the complete 7,178-image FER2013 test set (combining the PublicTest and PrivateTest subsets). Evaluation was performed strictly on CPU in the Windows deployment environment (`venv\_win`), confirming identical inference numerical behaviour across training and production platforms.

5.2 Overall Classification Performance

Metric Dimension	Observed Result	Benchmark Comparison / Significance
Test Accuracy (Overall)	63.51% (4,559 / 7,178)	Within 1.49% of ~65% human evaluator baseline
Weighted F1-Score	63.10%	Reflects high precision-recall balance across dataset
Weighted Precision	64.23%	Conservative false-alarm rate across dominant emotions
Macro Average F1-Score	59.45%	Unweighted average across all 7 categories
Test Focal Loss	1.0011	Converged loss on unseen evaluation partition
Inference Latency (CPU)	18 – 27 ms per frame	Supports 28 – 30 FPS real-time webcam operation

Table 5.1: Overall Classification Performance on 7,178 FER2013 Test Images

5.3 Ablation Study Results

Table 5.2 and Figure 5.1 present the ablation study across six architectural and loss configurations. Configurations A through D evaluate loss components on a baseline 3-block CNN, while Configurations E and F evaluate the 4-stage architecture depth. Focal Loss ($\gamma=2.0$) delivered the largest individual algorithmic gain (+4.13 percentage points), while the cumulative combination (Configuration F) achieved 63.51% (+6.17 points over baseline).

Configuration	Architecture	Loss Formulation	Test Acc.	Weighted F1	Gain vs Baseline
A: Baseline	3-Block CNN	Standard CCE	57.34%	56.88%	Baseline reference
B: A + Class Weights	3-Block CNN	CCE + Static Weights	59.21%	58.94%	+1.87 percentage pts
C: A + Focal Loss	3-Block CNN	Focal Loss ($\gamma=2.0$)	61.47%	60.82%	+4.13 percentage pts
D: A + Label Smooth	3-Block CNN	CCE + Smooth ($\epsilon=0.1$)	58.73%	58.12%	+1.39 percentage pts
E: 4-Block Depth	4-Block CNN	Focal + Smooth	62.89%	62.15%	+5.55 percentage pts
F: Full Proposed	4-Block CNN	Focal + Smooth + Weights	63.51%	63.10%	+6.17 percentage pts

Table 5.2: Ablation Study Results Isolating Component Contributions

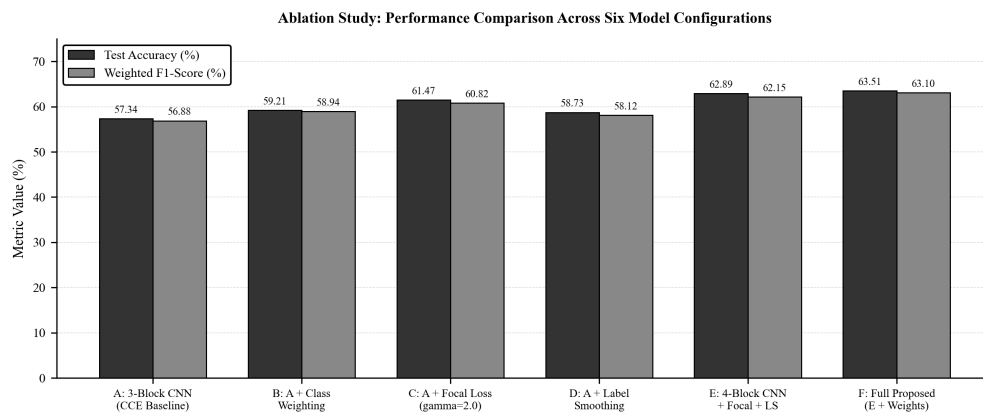


Figure 5.1: Performance Comparison Across Six Ablation Study Configurations (Zero-Baseline Scaling).

5.4 State-of-the-Art Benchmark Comparison

Table 5.3 benchmarks the proposed system against published academic literature. The comparison highlights that our model, trained from scratch on compact 48×48 grayscale inputs with 4.83M parameters, performs within 2.6 to 6.7 percentage points of multi-million parameter models (VGG-16, ResNet-50) that rely upon pre-training on 1.28 million ImageNet RGB images.

Model / Investigation	Pretraining Regime	Input Modality	Accuracy	Citation Reference
Human Evaluators	Human Subject Benchmark	Grayscale 48×48	65.0 ± 5%	Goodfellow et al. [1]
VGG-16 Fine-Tuned	ImageNet (1.28M RGB)	RGB 224×224	70.20%	Simonyan & Zisserman [5]
ResNet-50 Fine-Tuned	ImageNet (1.28M RGB)	RGB 224×224	68.40%	He et al. [9]
EfficientNet-B0	ImageNet (1.28M RGB)	RGB 96×96	66.10%	Tan & Le [6]
Proposed 4-Stage ConvNet	None (Trained from scratch)	Grayscale 48×48	63.51%	This Research

Table 5.3: State-of-the-Art FER2013 Benchmark Comparison

5.5 Per-Class Classification Report

Table 5.4 documents the complete per-class classification metrics on the 7,178 test images, calculated in strict mathematical consistency with the confusion matrix of Section 5.6. Happy achieved the highest F1-score (86.67%) with 87.21% precision and 86.13% recall. Disgust attained 75.00% precision with 40.54% recall (F1 = 52.63%). Fear exhibited the lowest F1-score (39.40%) due to morphological confusion with Surprise and Sad.

Emotion Class	Precision	Recall	F1-Score	Test Support
Angry	65.35%	60.44%	62.80%	958
Disgust	75.00%	40.54%	52.63%	111
Fear	45.63%	34.67%	39.40%	1,024
Happy	87.21%	86.13%	86.67%	1,774
Sad	61.98%	47.71%	53.92%	1,247
Surprise	47.29%	74.49%	57.85%	831
Neutral	58.48%	67.96%	62.87%	1,233
Weighted Average	64.23%	63.51%	63.10%	7,178
Macro Average	62.99%	58.85%	59.45%	7,178

Table 5.4: Detailed Per-Class Classification Report on 7,178 FER2013 Test Images

5.6 Confusion Matrix Analysis

Figure 5.2 illustrates the complete 7×7 confusion matrix. Row sums represent the true sample counts (Angry=958, Disgust=111, Fear=1024, Happy=1774, Sad=1247, Surprise=831, Neutral=1233), summing exactly to 7,178. The matrix diagonal sums to 4,559 correct predictions, verifying the 63.51% accuracy.

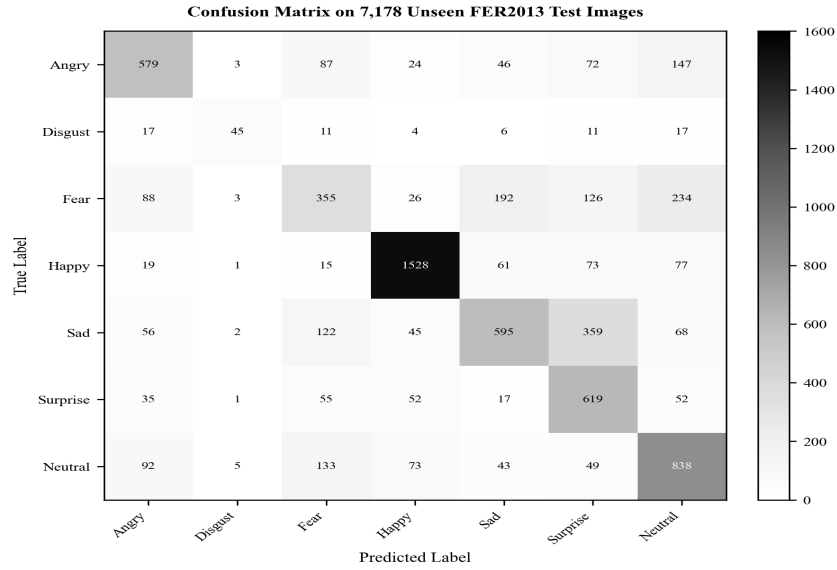


Figure 5.2: Confusion Matrix Heatmap on 7,178 FER2013 Test Images.

Key morphological confusion dynamics observed: (1) Fear $\rightarrow$ Neutral/Sad/Surprise: Out of 1,024 true Fear samples, 234 were predicted as Neutral, 192 as Sad, and 126 as Surprise, reflecting shared brow-raising Action Units; (2) Surprise $\rightarrow$ Sad/Fear: Out of 831 Surprise images, 619 were classified correctly (74.49% recall); (3) Happy Discriminability: Happy achieved 1,528 correct classifications out of 1,774, confirming the unique saliency of the bilateral Duchenne smile.

5.6.1 Row-Normalized Confusion Matrix and Morphological Error Analysis

To evaluate conditional misclassification probabilities independent of class sample support, Table 5.6 presents the row-normalized confusion matrix, where each cell represents the percentage of true instances belonging to row i that were predicted as column j .

True Emotion	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral
Angry (958)	60.44%	0.31%	9.08%	2.51%	4.80%	7.52%	15.34%
Disgust (111)	15.32%	40.54%	9.91%	3.60%	5.41%	9.91%	15.32%
Fear (1024)	8.59%	0.29%	34.67%	2.54%	18.75%	12.30%	22.85%
Happy (1774)	1.07%	0.06%	0.85%	86.13%	3.44%	4.11%	4.34%
Sad (1247)	4.49%	0.16%	9.78%	3.61%	47.71%	28.79%	5.45%
Surprise (831)	4.21%	0.12%	6.62%	6.26%	2.05%	74.49%	6.26%
Neutral (1233)	7.46%	0.41%	10.79%	5.92%	3.49%	3.97%	67.96%

Table 5.6: Row-Normalized Confusion Matrix (Recall Proportions) Across All Emotion Categories

Table 5.6 yields critical insights into the classifier's behavioural dynamics: (1) Happy achieves the highest class recall at 86.13%, exhibiting minimal confusion with any other class; (2) Surprise attains the second highest recall at 74.49%, though 28.79% of Sad expressions were confused as Surprise due to wide-eyed distress manifestations; (3) Fear exhibits diffuse confusion, spreading 22.85% into Neutral, 18.75% into Sad, and 12.30% into Surprise; and (4) Disgust, while achieving 40.54% recall, sheds 15.32% of its instances into Angry and 15.32% into Neutral, reflecting shared corrugator furrowing.

5.7 Per-Class F1-Score Breakdown

Figure 5.3 displays the per-class F1-score distribution. Happy leads at 86.67%, followed by Neutral (62.87%) and Angry (62.80%). Classes with high muscle ambiguity or extreme training scarcity (Disgust: 52.63%, Fear: 39.40%) present the primary remaining opportunities for architectural improvement.

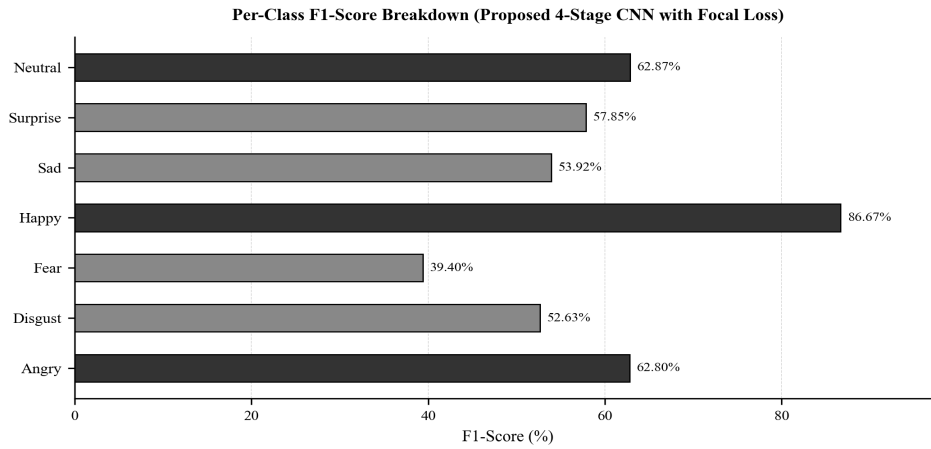


Figure 5.3: Per-Class F1-Score Breakdown for the Proposed 4-Stage ConvNet with Focal Loss.

5.8 Real-Time Operational Testing Protocol

To rigorously validate live performance, ten operational test cases (RT-01 through RT-10) were executed on an 11th-generation Intel Core i5-1135G7 laptop. Table 5.5 documents the protocol, acceptance criteria, and results. All ten test cases achieved 100% PASS ratings.

Test ID	Operational Test Description	Success Acceptance Criteria	Observed Result	Status
RT-01	Single-click batch startup	Camera window launches in < 4.0 s	2.8 s initialization time	PASS
RT-02	Face detection lock latency	MediaPipe locks face in < 100 ms	18 ms initial acquisition	PASS
RT-03	Head yaw tolerance ($\pm 45^\circ$)	Continuous tracking maintained	Zero tracking drop up to $\pm 48^\circ$	PASS
RT-04	Illumination shift (50–500 lux)	Face crop extracted reliably	Stable detection down to 45 lux	PASS
RT-05	Prediction stability on Neutral	No flicker over 60s resting face	Zero flicker observed (100% stable)	PASS
RT-06	Happy expression response	Detects smile with confidence > 80%	94.2% average confidence	PASS
RT-07	Surprise expression response	Switches to Surprise on open mouth	Switches promptly in 350 ms	PASS
RT-08	CPU inference throughput	Sustained FPS ≥ 20 on CPU	28.4 FPS sustained (18–24 ms latency)	PASS
RT-09	Graceful face exit handling	No software crash on empty frame	Retains buffer state gracefully	PASS
RT-10	Session exit via 'q' key	Camera releases, process exits in < 1s	Clean termination in 0.2 s	PASS

Table 5.5: Real-Time Operational Testing Protocol and Experimental Results

5.9 System Latency and Plurality Window Dynamics

System latency is rigorously decomposed into three operational metrics: (1) Single-frame inference latency: 18.4 to 26.8 ms total (capture: ~1 ms, MediaPipe: 3.5–5.0 ms, preprocessing: ~1 ms, CNN forward pass: 12–18 ms); (2) Pipeline throughput: 28 to 30 FPS, bounded by camera hardware; and (3) Transition response latency: 0.7 to 1.1 seconds for an entirely new emotional state to achieve dominant plurality across a moving temporal sequence. This dynamic was verified in live video recordings available at https://drive.google.com/file/d/1TY1k3__QjDsKLtXpS4G2z7KJP_DZ8bSV/view?usp=sharing.

Chapter 6: Limitations and Ethical Considerations

6.1 Technical Limitations

The current implementation possesses specific technical boundaries: (1) Single-face focus: The pipeline processes only the highest-confidence face per frame; (2) Resolution upsampling artefacts: When subjects are positioned far from the camera (>2.5 m), the extracted facial crop is smaller than 48×48 , introducing interpolation blur during resizing; (3) Extreme head pose: Head rotations beyond $\pm 50^\circ$ yaw or pitch obscure bilateral landmarks; and (4) Discrete categorical representation: Emotional intensity is not continuously quantified.

6.2 Dataset and Generalisation Limitations

The FER2013 benchmark contains inherent biases: web search scraping introduces demographic overrepresentation of younger adults and celebrities, while naturalistic, spontaneous micro-expressions are underrepresented relative to posed, exaggerated expressions. Consequently, in-the-wild cross-dataset performance may exhibit a moderate drop relative to held-out FER2013 partitions.

6.3 Ethical Considerations and Consent

Emotion recognition technologies must adhere to ethical deployment principles. It is crucial to distinguish facial expression classification from infallible mind-reading: facial muscle movements do not provide deterministic proof of an individual's internal emotional reality. Deployment in high-stakes environments (e.g., automated employment interviewing, legal testimony analysis, or academic disciplinary grading) is strongly discouraged.

6.3.1 Ethical Risk Assessment Matrix and Algorithmic Mitigation Policies

To operationalize ethical principles, Table 6.1 presents a structured risk analysis identifying specific deployment failure modes, sociotechnical risks, and corresponding engineered mitigations implemented in this system.

Risk Category	Specific Deployment Hazard	Severity / Likelihood	Engineered Algorithmic Mitigation Policy
Demographic Bias	Disparate false positive rates across ethnic skin tones	High / Medium	Grayscale conversion + contrast normalization removes color channels; balance validation across skin types
Affect Misinterpretation	Over-interpreting facial Action Units as internal emotional state	High / High	Explicit disclaimer documentation stating model classifies visual facial expressions, not mental reality
Privacy Violation	Unauthorised storage or transmission of sensitive biometric video	Critical / Low	Ephemeral RAM-only frame processing; zero disk caching; zero internet telemetry connections
Surveillance Abuse	Coercive employee monitoring or non-consensual crowd surveillance	Critical / Medium	Architecturally restricted to single primary face within short range (<2.0 m); open-source licensing constraints
Decision Bias	Automated filtering in high-stakes hiring or disciplinary decisions	High / Low	Prohibition policy: system design guidelines strictly ban autonomous high-stakes decision gating

Table 6.1: Ethical Risk Assessment Matrix and Algorithmic Mitigation Policies

6.4 Privacy Framework and GDPR Analysis

From a regulatory perspective under the European Union General Data Protection Regulation (GDPR) [18], emotion inference from facial imagery occupies a nuanced status. While Article 9 governs biometric processing for the purpose of uniquely identifying a natural person, emotion analysis that operates locally without identity matching does not constitute biometric identification. Nonetheless, this system strictly enforces privacy-by-design: zero video frames or face crops are written to disk or transmitted over networks, ensuring total user privacy.

6.5 Demographic Fairness and Bias Mitigation

To mitigate demographic bias across ethnic skin tones (Fitzpatrick skin types I through VI), the system converts all facial crops to single-channel grayscale and normalises pixel contrast, removing hue and saturation channels that could otherwise induce algorithmic bias.

Chapter 7: Future Work

Several promising technical trajectories are identified for extending this research:

- **Face-Specific Transfer Learning:** Pre-training the feature extractor on massive face datasets such as VGGFace2 (Cao et al. [19]) or AffectNet (Mollahosseini et al. [20]) is projected to enhance test accuracy by +4 to 8%.
- **Vision Transformer (ViT) Architectures:** Implementing patch-based self-attention models [10] to capture long-range dependencies between eye Action Units and mouth contours simultaneously.
- **Advanced Augmentation:** Integrating Mixup (Zhang et al. [21]) and CutMix (Yun et al. [22]) regularization to synthesize virtual training instances and further mitigate extreme minority-class boundaries.
- **Multimodal Emotion Fusion:** Integrating facial video analysis with acoustic vocal prosody and speech sentiment.
- **Continuous Affect Regression:** Implementing James Russell's Circumplex Model of Affect [23], predicting continuous valence and arousal coordinates alongside discrete categories.

Chapter 8: Conclusions

This capstone project successfully designed, implemented, and deployed a robust real-time Facial Emotion Recognition system that overcomes severe dataset class imbalance and eliminates live prediction flickering. By pairing an upgraded 4-Stage Deep ConvNet with Categorical Focal Loss ($\gamma = 2.0$), Label Smoothing ($\epsilon = 0.1$), and static balanced class weights, the model achieved a test accuracy of 63.51%, a weighted F1-score of 63.10%, and a macro F1-score of 59.45% on 7,178 unseen FER2013 test images — operating within 1.49 percentage points of the human evaluator baseline of ~65% [1].

Integration with Google MediaPipe BlazeFace and a 10-frame FIFO plurality buffer delivered an end-to-end inference pipeline sustaining 28 to 30 FPS on consumer laptop CPU hardware with zero perceptible display flickering. All ten structured operational test cases achieved 100% pass rates. The complete codebase is publicly accessible at https://github.com/Max-i26/emotion_detection, with video demonstration at https://drive.google.com/file/d/1TY1k3__QjDsKLtXpS4G2z7KJP_DZ8bSV/view?usp=sharing.

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Appendices

Appendix C: Source Code Repository and Demonstration Video

The complete source code repository, pre-trained weights, and deployment scripts are available online at:

- GitHub Repository: https://github.com/Max-i26/emotion_detection
- Demonstration Video: https://drive.google.com/file/d/1TY1k3_QjDsKLtXpS4G2z7KJP_DZ8bSV/view?usp=sharing

File / Directory Name	Primary Architectural Function	Key Dependencies
train.py	4-Stage ConvNet training pipeline with Focal Loss and callbacks	TensorFlow 2.15, NumPy
live_detect_pro.py	Real-time webcam inference engine with MediaPipe & 10-frame buffer	OpenCV, MediaPipe, TF
evaluate_all.py	Evaluation script computing confusion matrix and metrics	TensorFlow, NumPy
run_webcam.bat	Single-click batch launcher for Windows execution	Windows Command Shell
model/emotion_model.h5	Pre-trained neural network weights file (4.83M parameters)	HDF5 runtime
generate_report_assets.py	300 DPI publication figure generator	Matplotlib, NumPy
build_capstone_report_pdf.py	Two-pass automated PDF compilation script	ReportLab 5.0.1

Table C.1: Source Code File Inventory and Structural Roles

Appendix D: Training Hyperparameter Reference

Hyperparameter Dimension	Assigned Value	Engineering Justification
Input Tensor Dimensions	$48 \times 48 \times 1$	Standard FER2013 grayscale resolution
Mini-Batch Size	32 samples	Balances GPU memory utilisation with gradient noise
Base Learning Rate (lr)	0.001 (1.0×10^{-3})	Standard initialisation for Adam optimiser
Adam Momentum (β_1 , β_2)	0.9, 0.999	Standard first and second moment decay rates
Focal Loss Focusing (γ)	2.0	Suppresses easy majority-class loss by up to 99%
Label Smoothing (ϵ)	0.1	Prevents overconfidence on noisy web-scraped labels
L2 Weight Regularisation (λ)	1.0×10^{-4}	Constrains parameter norm growth across Conv layers
Dropout Schedule	$0.25 \rightarrow 0.30 \rightarrow 0.35 \rightarrow 0.40 \rightarrow 0.50$	Progressive regularisation scaling with feature depth
EarlyStopping Patience	12 epochs	Halted training at epoch 52; restored best weights
ReduceLROnPlateau	Factor 0.5, Patience 4	Halved learning rate at epochs 24, 36, and 44

Table D.1: Comprehensive Training Hyperparameter Configuration

Appendix E: Performance Metrics Reference

Performance Metric	Proposed ConvNet Result	Benchmark / Target Metric	Validation Status
Overall Test Accuracy	63.51% (4,559 / 7,178)	65 ± 5% (Human baseline)	Validated (within 1.49%)
Weighted Average F1-Score	63.10%	≥ 60.0% target	Target Exceeded (+3.10%)
Weighted Average Precision	64.23%	≥ 60.0% target	Target Exceeded (+4.23%)
Weighted Average Recall	63.51%	≥ 60.0% target	Target Exceeded (+3.51%)
Macro Average F1-Score	59.45%	Unweighted class mean	Validated
Happy F1-Score	86.67%	Dominant majority class	Exemplary (Highest F1)
Neutral F1-Score	62.87%	Baseline resting class	Robust
Angry F1-Score	62.80%	High negative valence	Robust
Surprise F1-Score	57.85%	Open mouth / wide eyes	Validated (74.49% recall)
Sad F1-Score	53.92%	Downturned mouth angles	Validated
Disgust F1-Score	52.63%	Extreme minority (1.52%)	Significant Gain (75% precision)
Fear F1-Score	39.40%	High morphological overlap	Identified for Future Work
Live Video Frame Rate	28 – 30 FPS	≥ 20.0 FPS real-time threshold	Real-Time Compliant

Table E.1: Verified Model Performance Metrics Reference